Installation Instructions

Upgrading Temperature Controller Software

September 25, 2025

BF1000-1234517327-115 Version 2.0, en-US
Original instructions

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Contact information

Bluefors Oy
Arinatie 10
00370 Helsinki
Finland
support@bluefors.com
+358 9 5617 4800

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1 Introduction

This information applies to upgrading the Bluefors Temperature Controller software. For detailed instructions about Bluefors Temperature Controller, refer to “Bluefors Temperature Controller User Manual”.

These instructions are essential for the use of the Bluefors product. For safe and proper use of the product, read the instructions before use. Keep them for future reference.

2 Preparing for the upgrade

2.1 Contents of the update package

The firmware update files are available on the Bluefors Support website. Notifications of available updates are sent by email.

NOTE

Although the update package is called a firmware update package, in the older Temperature Controller versions the version information is given as a software version or application software version. For more information, refer to “Checking the current software version”.

Figure 1: Firmware package available in Bluefors Support

The update package is called, for example, “Bluefors_TC-Firmware_Update_1.2.3_release”.

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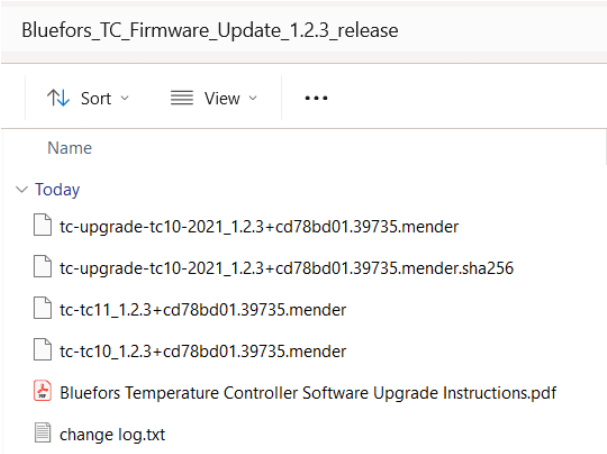

Figure 2: Contents of the upgrade package

The package contains these files.

Files starting with ”tc-upgrade-tc10-...” update the 0.20 versions. In this case, the USD drive may only contain the tc-upgrade-tc10-...” files.

If you have version 1.0.2, refer to the instructions in “Upgrading from 1.0.2”.

Files starting with ”tc-tc11_...” and ”tc-tc10_...” are used to upgrade versions 1.x.y onwards to the newest version. In this case, the USB drive may contain multiple files.

Temperature Controller has two hardware variants, 1.0 and 1.1, in which the main difference is the different power supply. For more information, refer to “Bluefors Temperature Controller User Manual”.

2.2 System requirements and specifications

- Operating system
  - Windows PC (Windows 11)
- USB drive specification
  - Capacity: 8–32 GB
  - File system: FAT32
  - Connector type: USB Type-A
  - The USB drive must be empty

NOTE

The USB drive must be formatted to FAT32 format before the update.

2.3 Formatting the USB drive to FAT32 format

NOTE

This method only works on USB flash drives that are maximum 32 GB in size.

To format the USB drive, do these steps.

1. Insert the USB drive into your computer's USB port.

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2. Select **Start** (Windows icon).
3. Open File Explorer.
4. Select **This PC**.
5. Right-click on your USB drive and select **Format**.

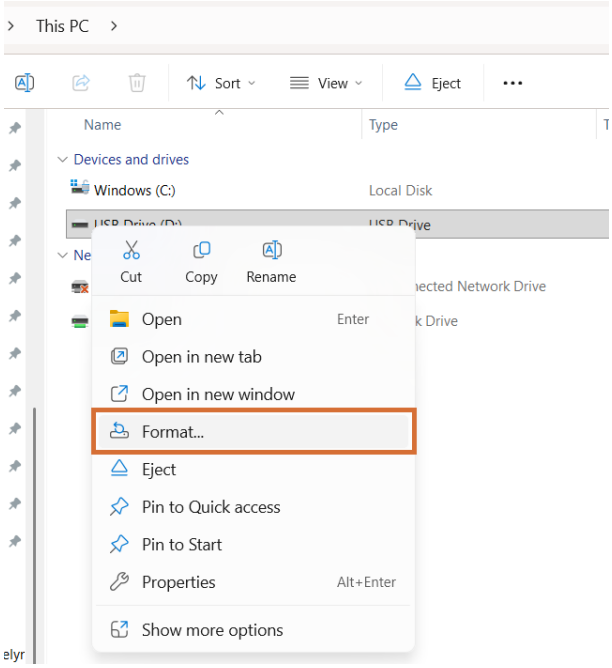

Figure 3: Format USB drive

6. From the **File system** drop-down list, select **FAT32** or **exFAT**.

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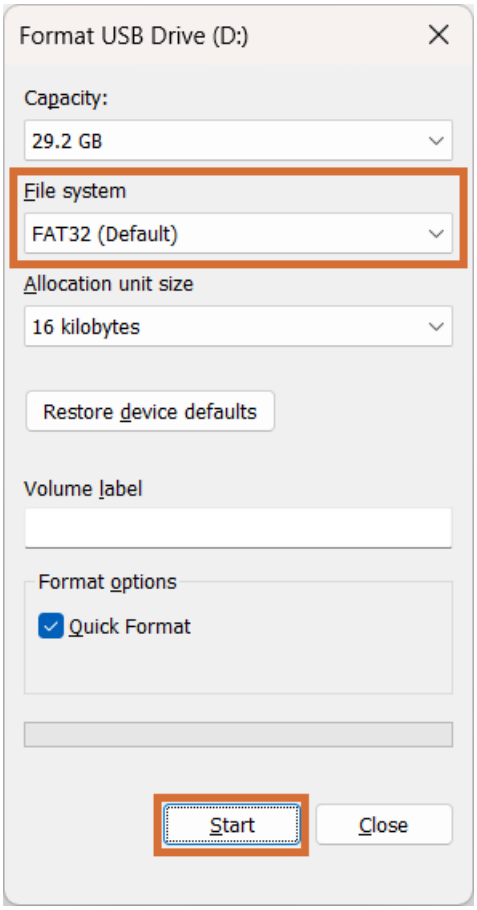

Figure 4: Select FAT32 from the File system drop-down list

7. Select **Start**.
8. To examine that the USB drive has been formatted, open **File Explorer**, right-click on your USB drive, and select **Properties**.
9. A window opens, and you should see that the file system is FAT32.

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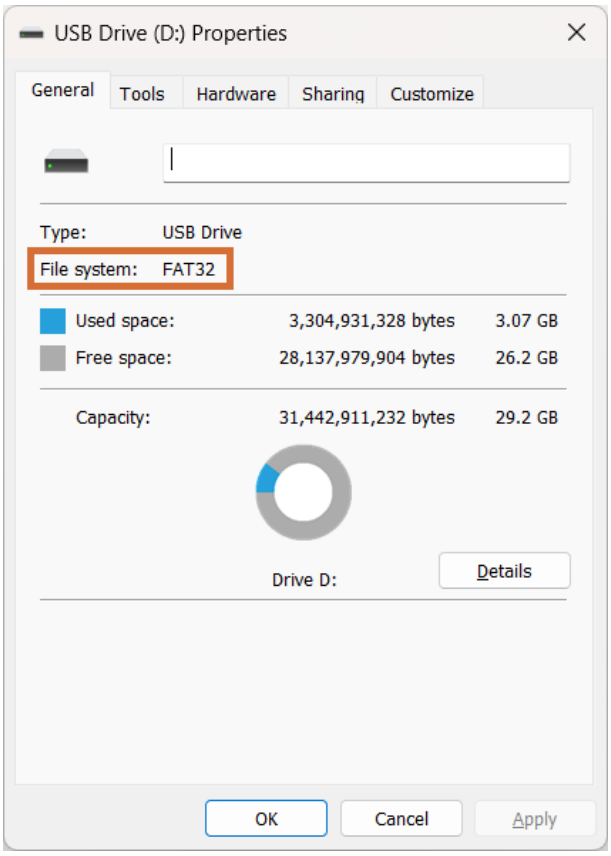

Figure 5: USB drive formatted in FAT32

2.4 Checking the current software version

To check your current **software version**, select ⓘ **Information** in the Temperature Controller home screen toolbar. The information on the screen unfortunately also contains the firmware version, which refers to the sub-firmware components, which in newer versions are known as the measurement interface.

These are the correct software versions that Bluefors has published for the customers.

- 0.20
- 1.0.2
- 1.1.0
- 1.1.2
- 1.2.0
- 1.2.3

Note that there is an incorrect version 0.3.11 which stands for the above-mentioned sub-firmware version.

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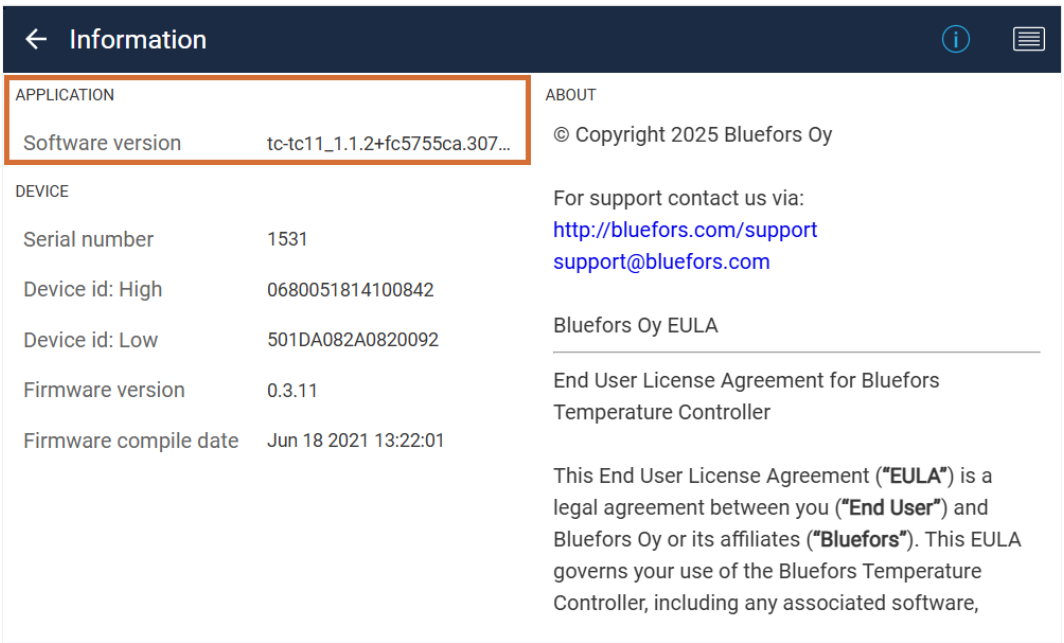

Figure 6: Check software version

3 Upgrading the device

The firmware update files are available on the Bluefors Support website. Notifications of available updates are sent by email.

Make sure that the USB drive has been formatted to FAT32 format. For instructions about formatting the USB drive, refer to “Formatting the USB drive to FAT32 format”.

3.1 Upgrading from 0.20 to 1.x.y

If you have Temperature Controller with an older version than 1.0.0, that is, 0.20, only copy these two files from the update file package to the USB drive.

- tc-upgrade-tc10-2021_[version]+[hash].mender
- tc-upgrade-tc10-2021_[version]+[hash].mender.sha256

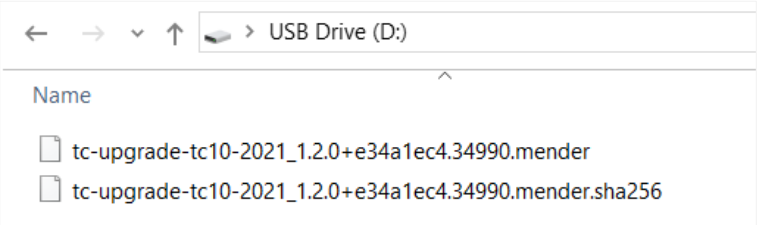

Figure 7: Upgrade files from 0.20 to 1.x

If you think that your version is 0.3.11, check the software version again as instructed in “Checking the current software version”.

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If the USB drive also contains some other files, the update may not start. If the USB drive has not been formatted to FAT32 format, the upgrade gets stuck and an error message "Looking for files" is displayed.

3.2 Upgrading from 1.0.2

If you have version 1.0.2 installed, only one file, depending on the version, must be present on the USB drive. The file is in this format:

- tc-tc10_[version]+[hash].mender

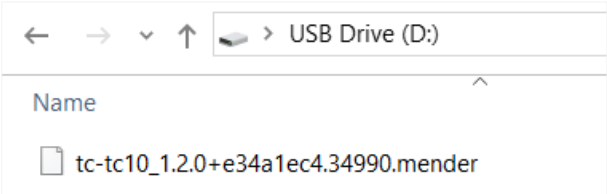

Figure 8: Upgrade file from 1.0.2

If the USB drive also contains some other files, the update may not start.

3.3 Updating the device manually

Currently Temperature Controller can only be updated manually with the USB drive.

To update the system manually, do these steps.

1. Format the USB drive to FAT32 format according to the instructions in “Formatting the USB drive to FAT32 format”.
2. Download the necessary update files from the Bluefors Support website. You can find the address in the **Information** view under **About**.
3. Connect the USB drive to the Windows PC.
4. Copy all necessary files to the USB drive. Copy only files, not folders, to the USB drive.

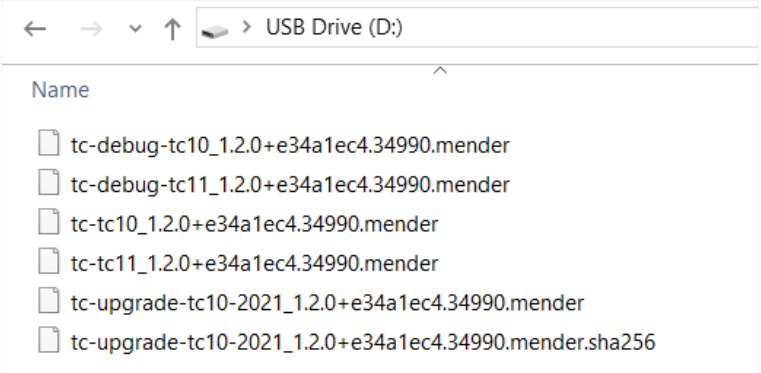

Figure 9: Update files

5. Safely remove the USB drive from the Windows PC. Right-click the USB drive and select **Eject**.

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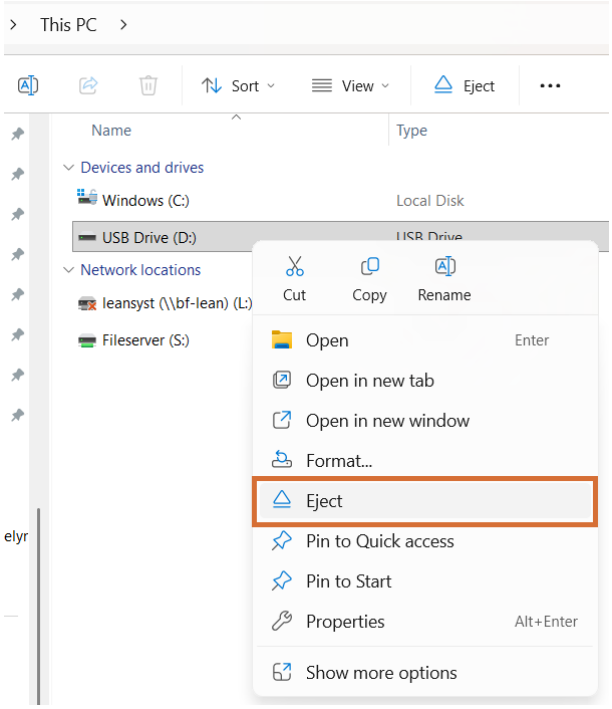

Figure 10: Eject USB drive

Figure 11: Windows notification to safely remove USB drive

6. Insert the USB drive to the **USB** (DEBUG) port on the back of the Temperature Controller control unit.

Figure 12: USB port on the back of the control unit

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7. In **System information**, to start the update, select **Update from USB**.

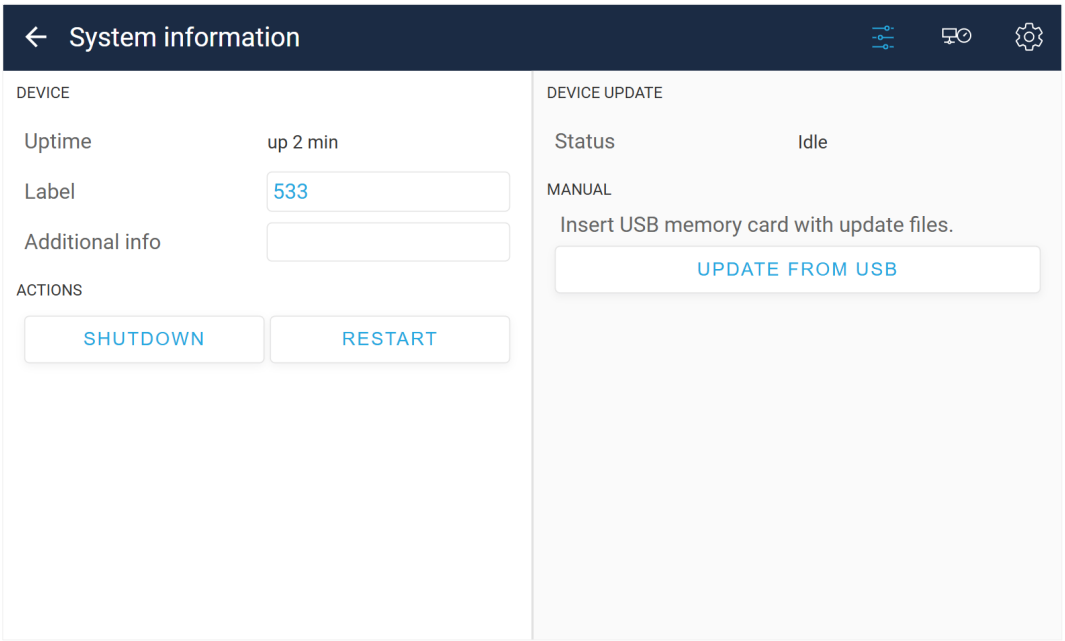

Figure 13: Update from USB

8. The device starts the update, and an information message is displayed on the screen. The update takes approximately 15 min. Select **OK**.

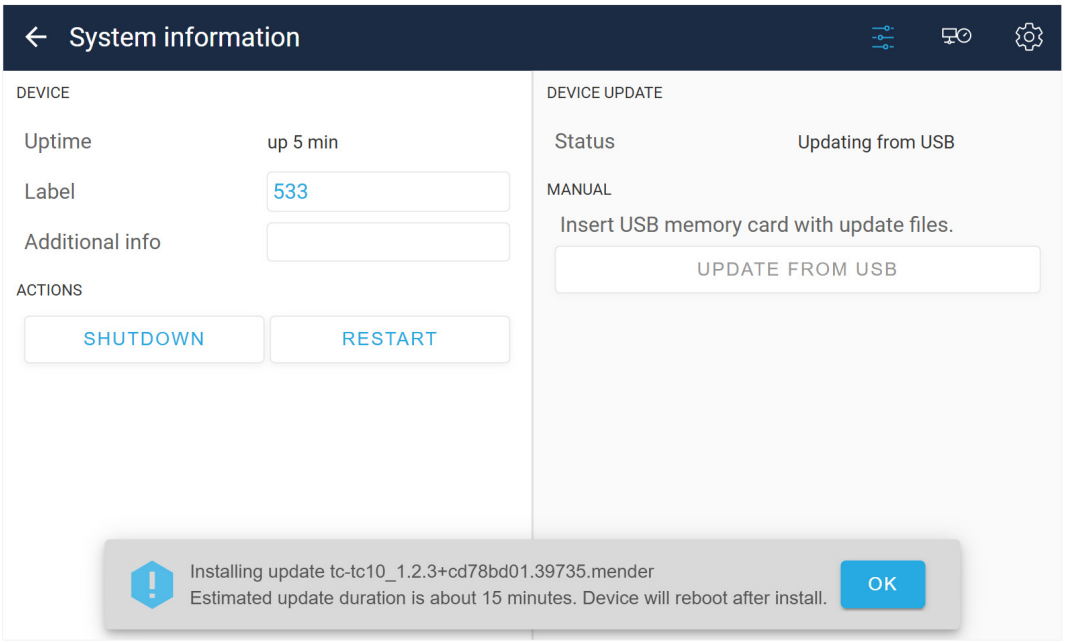

Figure 14: Update status in System information

9. When the update is ready, an information message about restart is displayed on the screen. Select **OK** and wait until the device has restarted.
10. After the restart, Temperature Controller opens on the Home screen, and the update is completed. Select **OK**.

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Figure 15: The update is completed

NOTE

Sometimes Temperature Controller restarts twice.

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